Compiled Solutions – Intermediate Division

1. 01-02 C1 Recursive Functions

$f(7) = f(5) + 2 = 7 + 2 = 9$
$f(5) = f(3) + 2 = 5 + 2 = 7$
$f(3) = f(1) + 2 = 3 + 2 = 5$
$f(1) = 1 + 2 = 3$

9

2. 01-02 C1 Recursive Functions

$f(7,4) = 4 - f(2,6) = 4 - 7 = -3$
$f(2,6) = 2 - f(0,5) = 2 - (-5) = 7$
$f(0,5) = 0 - 5 = -5$

–3

3. 02-03 C1 Recursive Functions

$F(8) = F(F(5)) + 4 = 11 + 4 = 15$
$F(5) = 7$
$F(7) = F(F(4)) + 4 = 7 + 4 = 11$
$F(4) = 5$

15

4. 02-03 C1 Recursive Functions

$F(1) = \frac{1}{2}$
$F(2) = 2/3 * F(1) + 1 = 2/3 * \frac{1}{2} + 1 = 4/3$
$F(3) = \frac{3}{4} * 4/3 + 1 = 2$
$F(4) = 4/5 * 2 + 1 = 13/5$
$F(5) = 5/6 * 13/5 + 1 = 19/6$
$F(6) = 6/7 * 19/6 + 1 = 26/7$

26/7 or 3 5/7

5. 03-04 C1 Recursive Functions

$f(11,7) = f(7 - 11, 11 - 1) + 3 = -14 + 3 = -11$
$f(-4, 10) = -4 - 10 = -14$

-14

6. 03-04 C1 Recursive Functions

$f(8) = f(f(5)) + 4 = f(2) + 4 = 1 + 4 = 5$
$f(5) = f(f(2)) + 4 = f(1) + 4 = -2 + 4 = 2$
$f(2) = 2^2 - 3 = 1$
$f(1) = 1^2 - 3 = -2$

5

7. 04-05 C1 Recursive Functions

f (12) = f (10) + 12 = 9 + 12 = 21 — 21
f (10) = f (7) – 10 = 19 – 10 = 9
f (7) = 19

8. 04-05 C1 Recursive Functions

f (7) = f (5) + 1 = 2 + 1 = 3 — 2
f (5) = f (3) + 1 = 1 + 1 = 2
f (3) = f (2) – 3 = 4 – 3 = 1
f (2) = 4
f (f (7)) = f (3) = 1
f (f (f (7))) = f (f (3)) = f (1) = 2

9. 05-06 C1 Recursive Functions

F(8,12) = F(9,10) + 3 = 136 — 136
F(9,10) = F(10,8) + 3 = 133
F(10,8) = F(7,10) – 1 = 130
F(7,10) = F(8,8) + 3 = 131
F(8,8) = 64 + 64 = 128

10. 05-06 C1 Recursive Functions

Stage 1 = 16 — Stage 5 = 1 — .25 or ¼
Stage 2 = 8 — Stage 6 = .5
Stage 3 = 4 — Stage 7 = .25
Stage 4 = 2

11. 06-07 C1 Recursive Functions

f(10) = f(5) – 3 = 7 — f(f(f(f(f(10))))) = f(f(f(f(7)))) = — 7
f(5) = f(7) + 2 = 10 — = f(f(f(8))) = f(f(5)) = f(10) = 7
f(7) = f(9) + 2 = 8
f(9) = f(4) – 3 = 6
f(4) = f(6) + 2 = 9
f(6) = f(8) + 2 = 7
f(8) = f(3) – 3 = 5
f(3) = 3 + 5 = 8

12. 06-07 C1 Recursive Functions

f(30,12) = max(f(26,15), f(12,30) = max(396,360) = 396 396
f(22,15) = max(f(22,18), f(15,26) = max(396,390) = 396
f(22,18) = max(f(18,21), f(18,22) = max(378,396) = 396
f(18,22) = 18 * 22 = 396
f(15,26) = 15*26 = 390
f(18,21) = 18*21 = 378
f(12*30) = 12* 30 = 360

13. 07-08 C1 Recursive Functions

$f\ (32) = f\ (32/2) - 1) + 3 = f\ (15) + 3 = 7 + 3 = 10$ 10
$f\ (15) = 2 * f\ (15\text{-}3) - 5 = 2 * f\ (12) - 5 = 2 * 6 - 5 = 7$
$f\ (12) = f\ (12/2 - 1) + 3 = f\ (5) + 3 = 3 + 3 = 6$
$f\ (5) = 2 * f\ (5 - 3) - 5 = 2 * f\ (2) - 5 = 2 * 4 - 5 = 3$
$f\ (2) = f\ (2/2 - 1) + 3 = f\ (0) + 3 = 1 + 3 = 4$
$f\ (0) = f\ (0/2 - 1)) + 3 = f\ (-1) + 3 = -2 + 3 = 1$
$f\ (-1) = (-1)^2 - 3 = 1 - 3 = -2$

14. 07-08 C1 Recursive Functions

$f\ (1) = 8 \Rightarrow f\ (2) = 2 * f\ (1) - 4 = 12$ 68
$f\ (2) = 12 \Rightarrow f\ (3) = 2 * f\ (2) - 4 = 20$
$f\ (3) = 20 \Rightarrow f\ (4) = 2 * f\ (3) - 4 = 36$
$f\ (4) = 36 \Rightarrow f\ (5) = 2 * f\ (4) - 4 = 68$

15. 08-09 C1 Recursive Functions

$f\ (35,8) = f\ (27,10) + 8 = 41$ 41
$f\ (27,10) = f\ (17,12) + 10 = 33$
$f\ (17,12) = f\ (5,14) + 12 = 23$
$f\ (5,14) = 5^2 - 14 = 11$

16. 08-09 C1 Recursive Functions

$f\ (1) = 1,\ f\ (2) = 6,\ f\ (3) = f\ (2) + 5 = 6 + 5 = 11,$ 235
$f\ (4) = f\ (3) + 5 = 11 + 5 = 16$

…
$f\ (10) = f\ (9) + 5 = 41 + 5 = 46$

The sum is : 1 + 6 + 11 + 16 + 21 +…+ 36 + 41 + 46 = 235
This is an arithmetic sequence with common difference of 5.
The formula for the sum is the sum of the first and last terms
times the number of terms divided by 2: 10*(1 +46)/2 = 235

17. 09-10 C1 Recursive Functions

$f(13) = 2 * f(10) - 7 = 2 * 19 - 7 = 31$ 31
$f(10) = f(9) + f(8) + 2 = 11 + 6 + 2 = 19$
$f(9) = f(8) + f(7) + 2 = 6 + 3 + 2 = 11$
$f(8) = f(7) + f(6) + 2 = 3 + 1 + 2 = 6$
$f(7) = f(6) + f(5) + 2 = 1 + 0 + 2 = 3$
$f(6) = 6 - 5 = 1$
$f(5) = 5 - 5 = 0$ Now substitute backwards.

18. 09-10 C1 Recursive Functions

$f(30) = f(29) + 4 = 119$ This is an arithmetic 119
$f(29) = f(28) + 4 = 115$ sequence with a common
: OR difference, d, of 4. The nth
$f(3) = f(2) + 4 = 11$ term can be calculated
$f(2) = 7$ using $f(n) = f(1) + (n-1)*d$.
$\therefore f(30) = 3 + 29 * 4$ Substitute backwards.

19. 10-11 C1 Recursive Functions

$f(18) = f(13) + 3 = 23$ 23
$f(13) = f(8) + 3 = 20$
$f(8) = f(15) - 2 = 17$
$f(15) = f(10) + 3 = 19$
$f(10) = f(5) + 3 = 16$
$f(5) = f(9) - 2 = 13$
$f(9) = 9 + 6 = 15$

20. 10-11 C1 Recursive Functions

$f(18) = f(13) - 2 = 6$ So $f(f(f(f(18))))$ -6
$f(13) = f(8) - 2 = 8$ $= f(f(f(6)))$
$f(8) = f(12) + 6 = 10$ $= f(f(2))$
$f(12) = f(7) - 2 = 4$ $= f(-2)$
$f(7) = f(11) + 6 = 6$ $= -6$
$f(11) = f(6) - 2 = 0$
$f(6) = 6 - 4 = 2$
$f(-2) = -2 - 4 = -6$

21. 1**1-12 C1 Recursive Functions**

$f(72) = f(36) + 72 = 72 + 72 = 144$ 144
$f(36) = f(18) + 36 = 36 + 36 = 72$
$f(18) = f(9) + 18 = 18 + 18 = 36$
$f(9) = 2*9 = 18$

22. 11-12 C1 Recursive Functions

$f(0) = f(2) + 1 = 4$
$f(2) = f(4) + 1 = 3$
$f(4) = f(6) + 1 = 2$
$f(6) = 1$
$f(3) = f(5) + 1 = 4$
$f(5) = f(7) + 1 = 3$ $f(7) = 2$

So $f(f(f(f(0))))$ 4
$= f(f(f(4)))$
$= f(f(2))$
$= f(3)$
$= 4$

23. 12-13 C1 Recursive Functions

$f(16) = f(f(14)) + 3 = f(1) + 3 = -3 + 3 = 0$ 0
$f(14) = f(f(12)) + 3 = f(2) + 3 = -2 + 3 = 1$
$f(12) = f(f(10) + 3 = f(3) + 3 = -1 + 3 = 2$
$f(10) = f(11) - 1 = 4 - 1 = 3$
$f(11) = f(f(9)) + 3 = f(5) + 3 = 1 + 3 = 4$
$f(9) = 9 - 4 = 5$ and $f(5) = 5 - 4 = 1$

24. 12-13 C1 Recursive Functions

$f(10,3) = f(8,4) + 10 = 30 + 10 = 40$ 40
$f(8,4) = f(6,5) + 8 = 22 + 8 = 30$
$f(6,5) = f(4,6) + 6 = 16 + 6 = 22$
$f(4,6) = f(1,8) + 6 = 10 + 6 = 16$
$f(1,8) = f(-2,10) + 8 = 2 + 8 = 10$
$f(-2,10) = (-2) * (-2) - 2 = 2$

25. 13-14 C1 Recursive Functions

$f(12) = 2 * f(9) - 3 = 2 * 49 - 3 = 95$ 95
$f(9) = 2 * f(6) - 3 = 2 * 26 - 3 = 49$
$f(6) = f(8) + 1 = 25 + 1 = 26$
$f(8) = 2 * f(5) - 3 = 2 * 14 - 3 = 25$
$f(5) = f(7) + 1 = 13 + 1 = 14$
$f(7) = 2 * f(4) - 3 = 2 * 8 - 3 = 13$
$f(4) = 4 + 4 = 8$

26. 13-14 C1 Recursive Functions

$f(10,2) = f(8,4) + 2 = 38 + 2 = 40$ 40
$f(8,4) = f(6,6) + 2 = 36 + 2 = 38$
$f(6,6) = f(7,5) - 1 = 37 - 1 = 36$
$f(7,5) = f(5,7) + 2 = 35 + 2 = 37$
$f(5,7) = 5 * 7 = 35$

27. 14-15 C1 Recursive Functions

$f(14) = f(12) - 1 = 9 - 1 =$ 8
$f(12) = f(10) - 1 = 10 - 1 = 9$
$f(10) = f(8) - 1 = 11 - 1 = 10$
$f(8) = f(7) + 4 = 7 + 4 = 11$
$f(7) = f(6) + 4 = 3 + 4 = 7$
$f(6) = 6 - 3 = 3$ $f(5) = 5 - 3 = 2$
$f(1) = 1 - 3 = -2$ $f(-3) = -3 - 3 -6$

28. 14-15 C1 Recursive Functions

$f(10) = f(7) - 1 = 8 - 1 = 7$ So $f(f(f(10)))$ 6
$f(7) = f(8) + 2 = 6 + 2 = 8$ $= f(f(7))$
$f(8) = f(5) - 1 = 7 - 1 = 6$ $= f(8)$
$f(5) = 5 + 2 = 7$ $= 6$

29. 15-16 C1 Recursive Functions

$f(1) = 2$ $f(2) = 0$ 24
$f(3) = 2 * f(2) + f(1) = 2 * 0 + 2 = 2$
$f(4) = 2 * f(3) + f(2) = 2 * 2 + 0 = 4$
$f(5) = 2 * f(4) + f(3) = 2 * 4 + 2 = 10$
$f(6) = 2 * f(5) + f(4) = 2 * 10 + 4 = 24$

30. 15-16 C1 Recursive Functions

$f(16) = f(12) + 2 = 11 + 2 = 13$ So $f(f(f(16)))$ 11
$f(12) = f(8) + 2 = 9 + 2 = 11$ $= f(f(13))$
$f(8) = f(9) - 1 = 10 - 1 = 9$ $= f(12)$
$f(9) = f(10) - 1 = 11 - 1 = 10$ $= 11$
$f(10) = f(6) + 2 = 9 + 2 = 11$
$f(6) = 6 + 3 = 9$
$f(13) = f(9) + 2 = 10 + 2 = 12$